

ity. However, above these temperatures these are rarely used due to contamination of platinum at higher temperatures. Advantages of RTDs include high accuracy, low drift and suitability for precision applications.

Two common effects that can cause errors in measurements with RTDs are self-heating and lead-wire effects. These are explained below:

Self-heating. Passing current through an RTD generates a voltage across the RTD. By measuring this voltage, its resistance and, thus, its temperature can be determined. This causes self-heating of the RTD, leading to change in its resistance. This, in turn, results in error in the measurement. Self-heating effects can be minimised by supplying lower excitation current.

Lead-wire resistance. An RTD is often connected to a measurement device with a two-wire connection. These two wires that provide the RTD with its excitation current are

also used to measure the voltage across the sensor. As RTDs have low nominal resistance, measurement accuracy can be greatly affected by lead-wire resistance. For example, one per cent measurement error can be caused by lead wires with a resistance of one ohm connected to a 100-ohm platinum RTD. To eliminate the effects of lead-wire resistance, a three-wire or four-wire connection method can be used, which creates high-impedance paths to effectively eliminate errors caused by lead-wire resistance.

Semiconductor temperature sensors

Semiconductor sensors are the easiest to measure temperature. These are designed to provide highly-accurate, repeatable results that require no compensation circuitry, lookup table or calibration. Modern silicon (IC) temperature sensors offer high accuracy and high linearity

over an operating range of about -55°C to $+150^{\circ}\text{C}$. The extremely low operating current minimises self-heating and maximises battery life. Often, these are integrated into multi-function ICs, which perform a number of other hardware monitoring functions. These are also useful in CJC circuits for wide-temperature-range thermocouples.

Semiconductor sensors can be local temperature sensors (analogue/digital) or remote temperature sensors. Local sensors are available with both analogue and digital outputs.

Analogue sensors. Until recently, all temperature sensors on the market provided analogue outputs. Unfortunately, these analogue outputs require a comparator, an ADC or an amplifier at their output to make these useful. Analogue temperature sensors develop an output voltage proportional to temperature with a nominal coefficient of $6.25\text{mV}/^{\circ}\text{C}$ and $10\text{mV}/^{\circ}\text{C}$, respectively.

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